THE DIRECT SUM OF THE TWISTS OF THE STRUCTURE SHEAF GENERATES THE DERIVED CATEGORY OF PROJECTIVE SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The right orthogonal of an object is stable under the triangulated subcategory it generates. invoked at proof lines 4386-4389. Extends the Hom vanishing from E to every object of the subcategory generated by E.
- On projective space the twists O_X(-m) for m >= n + 1 lie in the subcategory generated by E, being quasi-isomorphic to finite complexes built from summands of E, by the resolution lemma for projective space. invoked at proof lines 4390-4396. Provides the negative twists as test objects.
- For an ample invertible module, an object whose twisted cohomology groups against all powers vanish must be zero, by the ampleness detection lemma. invoked at proof lines 4404-4408. Concludes K = 0 from the vanishing of the Hom groups.

Source reference. Original definition 09SJ: sources/stacks-chow/tags.txt, source lines 12357–12357. Cited at statement lines 4379-4381 as the notion being verified; the verification is inside the counted span.

Source reference. Original lemma 09SK: sources/stacks-chow/tags.txt, source lines 12358–12358. Cited at proof lines 4387-4389 to extend the Hom vanishing; the induction and conclusion are inside the counted span.

Source reference. Original lemma 0BQS: sources/stacks-perfect/pinned-original-source.tex, source lines 4327–4327. Cited at proof lines 4390-4392 to build O_X(-n-1); the induction producing all twists is inside the counted span.

Source reference. Original lemma 0BQR: sources/stacks-perfect/pinned-original-source.tex, source lines 4285–4285. Cited at proof lines 4407-4408 to conclude K = 0; the twisted cohomology computation is inside the counted span.

Source reference. Original lemma 01Q2: sources/stacks-chow/tags.txt, source lines 2251–2251. Cited at proof lines 4408-4409 for the ampleness of O_X(1); the detection argument is inside the counted span.

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Lemma 0.2. *Let A be a ring. Let $X = \mathbf{P}^n_A$. Then*

$$E = \mathcal{O}_X \oplus \mathcal{O}_X(-1) \oplus \ldots \oplus \mathcal{O}_X(-n)$$

is a generator (Derived Categories, Definition 09SJ) of $D_{QCoh}(X)$.

Proof. Let $K \in D_{QCoh}(\mathcal{O}_X)$. Assume $\mathrm{Hom}(E, K[p]) = 0$ for all $p \in \mathbf{Z}$. We have to show that $K = 0$. By Derived Categories, Lemma 09SK we see that $\mathrm{Hom}(E', K[p])$ is zero for all $E' \in \langle E \rangle$ and $p \in \mathbf{Z}$. By Lemma 0BQS applied with $a = -n - 1$ we see that $\mathcal{O}_X(-n-1) \in \langle E \rangle$ because it is quasi-isomorphic to a finite complex whose terms are finite direct sums of summands of E. Repeating the argument with $a = -n-2$ we see that $\mathcal{O}_X(-n-2) \in \langle E \rangle$. Arguing by induction we find that $\mathcal{O}_X(-m) \in \langle E \rangle$ for all $m \geq 0$. Since

$$\mathrm{Hom}(\mathcal{O}_X(-m), K[p]) = H^p(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_X(m)) = H^p(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_X(1)^{\otimes m})$$

we conclude that $K = 0$ by Lemma 0BQR. (This also uses that $\mathcal{O}_X(1)$ is an ample invertible sheaf on X which follows from Properties, Lemma 01Q2.) $\square$